PADHAKU
Ganita Manjari · Class 9 Mathematics



CHAPTER 2

Introduction to Linear Polynomials

NCERT Exercise Solutions

Think and Reflect · 2.1 Introduction	4 Q
Exercise Set 2.1	5 Q
Think and Reflect · 2.2 Linear Polynomials	3 Q
Exercise Set 2.2	7 Q
Think and Reflect · 2.3 Exploring linear patterns	4 Q
Exercise Set 2.3	5 Q
Think and Reflect · 2.4 Linear growth and linear decay	2 Q
Exercise Set 2.4	4 Q
Think and Reflect · 2.5 Linear Relationships	1 Q
Exercise Set 2.5	3 Q
Think and Reflect · 2.6 Visualising linear relationships	8 Q

✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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🤔 Think and Reflect · 2.1 Introduction

1 (Ex. 2) **Can you identify the terms, variables and coefficients of the algebraic expression 200l + 160w + 50lw?**

✔ ANSWER

- Terms: **200l, 160w and 50lw**
- Variables: **l and w**
- Coefficients: **200 (of l), 160 (of w) and 50 (of lw)**

2 (Ex. 2) **How is it different from the algebraic expression in Example 1 (4x + 5y + 3)?**

✔ ANSWER

- 4x + 5y + 3 has a constant term (3); 200l + 160w + 50lw has **no constant term**.
- In 200l + 160w + 50lw, two variables are multiplied together in the term **50lw**. In 4x + 5y + 3, no two variables are multiplied.

1 (Ex. 3) **Can you identify the terms, variables and coefficients of the algebraic expression $10x - x^2$?**

✔ ANSWER

- Terms: **$10x$ and $-x^2$**
- Variable: **x** (only one)
- Coefficients: **10 (of x) and −1 (of x^2)**

2 (Ex. 3) **Can you point out any similarity or difference between the algebraic expressions obtained in Examples 1 and 3?**

✔ ANSWER

- Similarity: both are algebraic expressions made of terms with numbers (coefficients) and variables.
- Difference: 4x + 5y + 3 has **two variables** (x, y), the highest power is 1, and it has a constant term.
- $10x - x^2$ has **only one variable** (x), it has a power 2 (x^2), and it has no constant term.

📝 Exercise Set 2.1

1 **Find the degrees of the following polynomials:**

(i) $2x^2 - 5x + 3$

(ii) $y^3 + 2y - 1$

(iii) -9

(iv) $4z - 3$

✔ ANSWER

Degree = highest power of the variable.

(i) Highest power of x is 2 → **degree 2**

(ii) Highest power of y is 3 → **degree 3**

(iii) $-9 = -9x^0$ → **degree 0** (constant polynomial)

(iv) Highest power of z is 1 → **degree 1**

2 **Write polynomials of degrees 1, 2 and 3.**

💡 SAMPLE ANSWER

Examples (many answers are possible):

- Degree 1: **$2x + 5$**
- Degree 2: **$x^2 - 3x + 1$**
- Degree 3: **$4y^3 + y^2 - 7$**

3 **What are the coefficients of x^2 and x^3 in the polynomial $x^4 - 3x^3 + 6x^2 - 2x + 7$?**

✔ ANSWER

Coefficient of x^2 = **6**

Coefficient of x^3 = **−3**

4 **What is the coefficient of z in the polynomial $4z^3 + 5z^2 - 11$?**

✔ ANSWER

There is no z term, so we can write it as $0 \times z$.

Coefficient of z = **0**

5 **What is the constant term of the polynomial $9x^3 + 5x^2 - 8x - 10$?**

✔ ANSWER

Constant term = **−10**

🤔 Think and Reflect · 2.2 Linear Polynomials

1 **Find the perimeter of squares with sides 1 cm, 1.5 cm, 2 cm, 2.5 cm and 3 cm. What will happen to the perimeters if the sides increase by 0.5 cm?**

✔ ANSWER

Perimeter = 4 × side

Side (cm)	1	1.5	2	2.5	3
Perimeter (cm)	4	6	8	10	12

When the side increases by 0.5 cm, the perimeter increases by 4 × 0.5 = **2 cm** each time.

2 **Chess club: amount paid = ₹(200 + 50m). If a player paid ₹750, how many matches did he play?**

✔ ANSWER

200 + 50m = 750

50m = 550

m = 11

He played **11 matches**.

3 **In Example 3, the area of the rectangle, $10x - x^2$, is a function of x. Can you interpret this as an input-output process? What value does the expression take when x = 6 cm?**

✔ ANSWER

Yes. Input: the length x. The machine works out $10x - x^2$. Output: the area.

For x = 6: $10 \times 6 - 6^2 = 60 - 36 = 24$

Area = **24 cm²** (length 6 cm, width 4 cm).

📝 Exercise Set 2.2

1 **Find the value of the linear polynomial 5x − 3 if:**
(i) x = 0 (ii) x = −1 (iii) x = 2

✔ ANSWER

(i) $5 \times 0 - 3 = \mathbf{-3}$

(ii) $5 \times (-1) - 3 = -5 - 3 = \mathbf{-8}$

(iii) $5 \times 2 - 3 = 10 - 3 = \mathbf{7}$

2 **Find the value of the quadratic polynomial $7s^2 - 4s + 6$ if: (i) $s = 0$ (ii) $s = -3$ (iii) $s = 4$**

✔ ANSWER

(i) $7 \times 0 - 0 + 6 = \mathbf{6}$

(ii) $7 \times 9 - 4 \times (-3) + 6 = 63 + 12 + 6 = \mathbf{81}$

(iii) $7 \times 16 - 4 \times 4 + 6 = 112 - 16 + 6 = \mathbf{102}$

3 **The present age of Salil's mother is three times Salil's present age. After 5 years, their ages will add up to 70 years. Find their present ages.**

✔ ANSWER

Let Salil's age = x years. Mother's age = 3x years.

After 5 years: $(x + 5) + (3x + 5) = 70$

$4x + 10 = 70$

$4x = 60$, so $x = 15$

Salil: **15 years**, mother: $3 \times 15 =$ **45 years**.

4 **The difference between two positive integers is 63. The ratio of the two integers is 2:5. Find the two integers.**

✔ ANSWER

Let the integers be 2k and 5k.

$5k - 2k = 63$

$3k = 63$, so $k = 21$

Integers: $2 \times 21 =$ **42** and $5 \times 21 =$ **105**.

5 **Ruby has 3 times as many two-rupee coins as she has five-rupee coins. If she has a total ₹88, how many coins does she have of each type?**

✔ ANSWER

Let five-rupee coins = x. Two-rupee coins = 3x.

$5 \times x + 2 \times 3x = 88$

$11x = 88$, so $x = 8$

8 five-rupee coins and $3 \times 8 =$ **24 two-rupee coins**.

Check: 40 + 48 = 88 ✓

6 **A farmer cuts a 300 feet fence into two pieces of different sizes. The longer piece is four times as long as the shorter piece. How long are the two pieces?**

✔ ANSWER

Let the shorter piece = x feet. Longer piece = 4x feet.

x + 4x = 300

5x = 300, so x = 60

Shorter piece: **60 feet**, longer piece: **240 feet**.

7 **If the length of a rectangle is three more than twice its width and its perimeter is 24 cm, what are the dimensions of the rectangle?**

✔ ANSWER

Let width = w cm. Length = (2w + 3) cm.

Perimeter: 2(length + width) = 24

2(2w + 3 + w) = 24

3w + 3 = 12

3w = 9, so w = 3

Width = **3 cm**, length = 2 × 3 + 3 = **9 cm**.

🤔 Think and Reflect · 2.3 Exploring linear patterns

1 **Predict the number of squares in the next three stages of the pattern (Fig. 2.4: 1, 3, 5, 7 tiles) and write the sequence of numbers up to Stage 7.**

✔ ANSWER

Each stage adds 2 tiles.

Stage 5: **9**, Stage 6: **11**, Stage 7: **13**

Sequence: **1, 3, 5, 7, 9, 11, 13**

2 **Using the expression 2n − 1, can you find out how many tiles will be there in the 15th stage and the 26th stage of the pattern? Also, which stage will contain 21 tiles and 47 tiles?**

✔ ANSWER

15th stage: 2 × 15 − 1 = **29 tiles**

26th stage: 2 × 26 − 1 = **51 tiles**

21 tiles: 2n − 1 = 21 → 2n = 22 → n = 11 → **Stage 11**

47 tiles: 2n − 1 = 47 → 2n = 48 → n = 24 → **Stage 24**

3 **Bela: amount left on the nth day = ₹(100 − 5n). What amount will be left on the 15th day? How many days will it take for the entire amount to be spent?**

✔ ANSWER

15th day: 100 − 5 × 15 = 100 − 75 = **₹25**

All spent when 100 − 5n = 0 → 5n = 100 → n = 20

It will take **20 days**.

4 **Auto-rickshaw fare = ₹(15n − 5) for n ≥ 2 km. For how many km will the fare be ₹130?**

✔ ANSWER

15n − 5 = 130

15n = 135

n = 9

The fare is ₹130 for **9 km**. (Check: 25 + 15 × 7 = 130 ✓)

Exercise Set 2.3

1 **A student has ₹500 in her savings bank account. She gets ₹150 every month as pocket money. How much money will she have at the end of every month from the second month onwards? Find a linear expression to represent the amount she will have in the nth month.**

✔ ANSWER

She adds ₹150 every month (assuming she saves all of it).

Month	1	2	3	4	5
Amount (₹)	650	800	950	1100	1250

Each month the amount increases by ₹150.

Amount in the nth month = **₹(500 + 150n)**

2 **A rally starts with 120 members. Each hour, 9 members drop out of the group. How many members will remain after 1, 2, 3, ... hours? Find a linear expression to represent the number of members at the end of the nth hour.**

✔ ANSWER

After 1 hour: 120 − 9 = 111

After 2 hours: 120 − 18 = 102

After 3 hours: 120 − 27 = 93, and so on.

Members: **111, 102, 93, 84, ...**

At the end of the nth hour: **120 − 9n**

3 **Suppose the length of a rectangle is 13 cm. Find the area if the breadth is (i) 12 cm, (ii) 10 cm, (iii) 8 cm. Find the linear pattern representing the area of the rectangle.**

✔ ANSWER

Area = length × breadth = 13 × b

(i) 13 × 12 = **156 cm²**

(ii) 13 × 10 = **130 cm²**

(iii) 13 × 8 = **104 cm²**

Linear pattern: **Area = 13b**. Each time the breadth decreases by 2 cm, the area decreases by 26 cm² (156, 130, 104, ...).

4 **Suppose the length of a rectangular box is 7 cm and breadth is 11 cm. Find the volume if the height is (i) 5 cm, (ii) 9 cm, (iii) 13 cm. Find the linear pattern representing the volume of the rectangular box.**

✔ ANSWER

Volume = 7 × 11 × h = 77h

(i) 77 × 5 = **385 cm³**

(ii) 77 × 9 = **693 cm³**

(iii) 77 × 13 = **1001 cm³**

Linear pattern: **Volume = 77h**. Each time the height increases by 4 cm, the volume increases by 308 cm³ (385, 693, 1001, ...).

5 **Sarita is reading a book of 500 pages. She reads 20 pages every day. How many pages will be left after 15 days? Express this as a linear pattern.**

✔ ANSWER

Pages left after n days = 500 − 20n

After 15 days: 500 − 20 × 15 = 500 − 300 = **200 pages**

Linear pattern: 480, 460, 440, … i.e. **500 − 20n** (decreases by 20 every day).

🤔 Think and Reflect · 2.4 Linear growth and linear decay

1 **C(d) = 100 + 60d. What is the cost for travelling 15 km? For how many kilometres will the cost of the journey be ₹700?**

✔ ANSWER

C(15) = 100 + 60 × 15 = 100 + 900 = **₹1000**

100 + 60d = 700 → 60d = 600 → d = 10

The cost is ₹700 for **10 km**.

2 **h(t) = 3 − 0.5t. What will be the height of the water at the end of 5 months?**

✔ ANSWER

h(5) = 3 − 0.5 × 5 = 3 − 2.5 = **0.5 m**

📝 Exercise Set 2.4

1 **Suppose a plant has height 1.75 feet and it grows by 0.5 feet each month.**

(i) Find the height after 7 months.

(ii) Make a table of values for t varying from 0 to 10 months and show how the height, h, increases every month.

(iii) Find an expression that relates h and t, and explain why it represents linear growth.

✔ ANSWER

(i) Height = 1.75 + 0.5 × 7 = 1.75 + 3.5 = **5.25 feet**

(ii)

t (months)	0	1	2	3	4	5	6	7	8	9	10
h (feet)	1.75	2.25	2.75	3.25	3.75	4.25	4.75	5.25	5.75	6.25	6.75

(iii) **h = 1.75 + 0.5t**. Every month the height increases by the same amount (0.5 feet), so it is linear growth.

2 **A mobile phone is bought for ₹10,000. Its value decreases by ₹800 every year.**

(i) Find the value of the phone after 3 years.

(ii) Make a table of values for t varying from 0 to 8 years and show how the value of the phone, v, depreciates with time.

(iii) Find an expression that relates v and t, and explain why it represents linear decay.

✔ ANSWER

(i) Value = 10000 − 800 × 3 = 10000 − 2400 = **₹7600**

(ii)

t (years)	0	1	2	3	4	5	6	7	8
v (₹)	10000	9200	8400	7600	6800	6000	5200	4400	3600

(iii) **v = 10000 − 800t**. Every year the value decreases by the same amount (₹800), so it is linear decay.

3 **The initial population of a village is 750. Every year, 50 people move from a nearby city to the village.**

(i) Find the population of the village after 6 years.

(ii) Make a table of values for t varying from 0 to 10 years and show how the population, P, increases every year.

(iii) Find an expression that relates P and t, and explain why it represents linear growth.

✔ ANSWER

(i) Population = 750 + 50 × 6 = 750 + 300 = **1050**

(ii)

t (years)	0	1	2	3	4	5	6	7	8	9	10
P	750	800	850	900	950	1000	1050	1100	1150	1200	1250

(iii) **P = 750 + 50t**. Every year the population increases by the same number (50), so it is linear growth.

4 **A telecom company charges ₹600 for a certain recharge scheme. This prepaid balance is reduced by ₹15 each day after the recharge.**

(i) Write an equation that models the remaining balance b(x) after using the scheme for x days. Explain why it represents linear decay.

(ii) After how many days will the balance run out?

(iii) Make a table of values for x varying from 1 to 10 days and show how the balance b(x) reduces with time.

✔ ANSWER

(i) **b(x) = 600 − 15x**. The balance decreases by the same amount (₹15) every day, so it is linear decay.

(ii) 600 − 15x = 0 → 15x = 600 → x = 40. The balance runs out after **40 days**.

(iii)

x (days)	1	2	3	4	5	6	7	8	9	10
b(x) (₹)	585	570	555	540	525	510	495	480	465	450

🤔 Think and Reflect · 2.5 Linear Relationships

1 **Can you guess what the numbers 20 and 150 in the equation y = 20x + 150 represent?**

✔ ANSWER

- **20** = the cost (₹) for each GB of data used.
- **150** = the fixed monthly fee (₹), paid even if no data is used (x = 0).

📝 Exercise Set 2.5

1 **A learning platform charges a fixed monthly fee and an additional cost per module. For 10 modules the bill was ₹400; for 14 modules it was ₹500. If y = ax + b, find a and b.**

✔ ANSWER

400 = 10a + b ... (1)

500 = 14a + b ... (2)

(2) − (1): 4a = 100, so a = 25

b = 400 − 10 × 25 = 150

a = 25, b = 150, so y = 25x + 150 (₹25 per module, fixed fee ₹150).

2 **A gym charges a fixed monthly fee and an additional cost per hour of badminton court. For 10 hours the bill was ₹800; for 15 hours it was ₹1100. If y = ax + b, find a and b.**

✔ ANSWER

800 = 10a + b ... (1)

1100 = 15a + b ... (2)

(2) − (1): 5a = 300, so a = 60

b = 800 − 10 × 60 = 200

a = 60, b = 200, so y = 60x + 200 (₹60 per hour, fixed fee ₹200).

3 **°C = a °F + b. Find a and b, given that ice melts at 0 °C and 32 °F, and water boils at 100 °C and 212 °F.**

✔ ANSWER

0 = 32a + b ... (1)

100 = 212a + b ... (2)

(2) − (1): 180a = 100, so a = 100/180 = 5/9

b = −32a = −32 × 5/9 = −160/9

a = 5/9, b = −160/9

So °C = (5/9) °F − 160/9, i.e. **°C = 5/9 (°F − 32)**.

🤔 Think and Reflect · 2.6 Visualising linear relationships

1 **Identify other points on the line y = 2x + 1 by completing the table.**

x	1	2	5	7	9	12	20
y	3			15			

✔ ANSWER

y = 2x + 1

x	1	2	5	7	9	12	20
y	3	5	11	15	19	25	41

Points: (1, 3), (2, 5), (5, 11), (7, 15), (9, 19), (12, 25), (20, 41).

Ex. 12 **Points (−1, −3), (0, 0), (1, 3), (3, 9), (4, 12) lie on a straight line. Can you guess its equation?**

✔ ANSWER

In each point, y is 3 times x.

Equation: **y = 3x**

Ex. 13 **Points (−3, 6), (−2, 4), (0, 0), (1, −2), (2, −4), (3, −6) lie on a straight line. Can you guess its equation?**

✔ ANSWER

In each point, y is −2 times x (e.g. 1 → −2, −3 → 6).

Equation: **y = −2x**

Ex. 14 **To graph y = ½x, we could take the points (0, 0) and (4, 2). Can you verify that these lie on the line?**

✔ ANSWER

(0, 0): ½ × 0 = 0 = y ✓

(4, 2): ½ × 4 = 2 = y ✓

Both points lie on y = ½x.

Fig. 2.9 **Does Fig. 2.9 (y = 2x, y = x, y = ½x) help you conclude anything about y = ax, a > 0, as a varies? What happens when a > 1 and when a < 1?**

✔ ANSWER

- All lines y = ax pass through the origin (0, 0) and rise from left to right.
- When **a > 1**, the line is **steeper** than y = x.
- When **a < 1**, the line is **less steep** than y = x.
- a is the slope of the line.

Fig. 2.11 **Does Fig. 2.11 (y = −⅓x, y = −x, y = −3x) help you conclude anything about y = −ax, a > 0, as a varies? What will happen when a > 1 and when a < 1?**

✔ ANSWER

- All lines y = −ax pass through the origin (0, 0) and go down from left to right (negative slope).
- When **a > 1**, the line is **steeper** than y = −x.

- When **a < 1**, the line is **less steep** than y = −x.

2 **Differentiate between the graphs of the equations y = 3x + 1 and y = −3x + 1.**

✔ ANSWER

- Both lines cut the y-axis at the same point **(0, 1)** (y-intercept 1).
- y = 3x + 1 has slope **3**: it rises from left to right (growth).
- y = −3x + 1 has slope **−3**: it falls from left to right (decay).
- Both are equally steep; each is the mirror image of the other in the y-axis.

3 **Graphs of y = 2x − 1, y = 2x + 1, y = 2x + 5: does this help you to conclude anything about y = ax + b when a is fixed but b varies?**

✔ ANSWER

- The slope a stays the same, so the lines are **parallel**.
- Each line cuts the y-axis at (0, b): changing b only **shifts the line up or down**.

📝 Exercise Set 2.6

1 (i) **Draw the graphs of y = 4x, y = 2x, y = x. Reflect on the role of 'a' and 'b'.**

✔ ANSWER

Points: y = 4x: (0, 0), (1, 4); y = 2x: (0, 0), (1, 2); y = x: (0, 0), (1, 1). Join each pair with a ruler.

- b = 0, so all three lines **pass through the origin**.
- a is positive, so all rise from left to right; **bigger a → steeper line** (y = 4x is steepest).

1 (ii) **Draw the graphs of y = −6x, y = −3x, y = −x. Reflect on the role of 'a' and 'b'.**

✔ ANSWER

Points: y = −6x: (0, 0), (1, −6); y = −3x: (0, 0), (1, −3); y = −x: (0, 0), (1, −1).

- b = 0, so all lines **pass through the origin**.

- a is negative, so all fall from left to right; the line with a = −6 is the **steepest**, y = −x the least steep.

1 (iii) **Draw the graphs of y = 5x, y = −5x. Reflect on the role of 'a' and 'b'.**

✔ ANSWER

Points: y = 5x: (0, 0), (1, 5); y = −5x: (0, 0), (1, −5).

- Both pass through the origin (b = 0) and are **equally steep**.
- y = 5x rises (a = 5 > 0); y = −5x falls (a = −5 < 0). They are mirror images of each other in the y-axis.

1 (iv) **Draw the graphs of y = 3x − 1, y = 3x, y = 3x + 1. Reflect on the role of 'a' and 'b'.**

✔ ANSWER

Points: y = 3x − 1: (0, −1), (1, 2); y = 3x: (0, 0), (1, 3); y = 3x + 1: (0, 1), (1, 4).

- Same a = 3, so the lines are **parallel**.
- b changes (−1, 0, 1): the lines cut the y-axis at (0, −1), (0, 0), (0, 1); b **shifts the line up or down**.

1 (v) **Draw the graphs of y = −2x − 3, y = −2x, y = 2x + 3. Reflect on the role of 'a' and 'b'.**

✔ ANSWER

Points: y = −2x − 3: (0, −3), (−1, −1); y = −2x: (0, 0), (1, −2); y = 2x + 3: (0, 3), (−1, 1).

- y = −2x − 3 and y = −2x have the same a = −2, so they are **parallel** (both fall); b shifts the line (y-intercepts −3 and 0).
- y = 2x + 3 has a = 2, so it rises and is **not parallel** to the other two; it cuts the y-axis at (0, 3).

End-of-Chapter Exercises

1 **Write a polynomial of degree 3 in the variable x, in which the coefficient of the x^2 term is −7.**

💡 SAMPLE ANSWER

One possible answer: **$x^3 - 7x^2 + 2x + 5$**

(Any polynomial with x^3 as the highest power and $-7x^2$ as the x^2 term is correct.)

2 **Find the values of the following polynomials at the indicated values of the variables.**

(i) $5x^2 - 3x + 7$ if $x = 1$

(ii) $4t^3 - t^2 + 6$ if $t = a$

✔ ANSWER

(i) $5 \times 1 - 3 \times 1 + 7 = 5 - 3 + 7 =$ **9**

(ii) Put $t = a$: **$4a^3 - a^2 + 6$**

3 **If we multiply a number by 5/2 and add 2/3 to the product, we get −7/12. Find the number.**

✔ ANSWER

Let the number be x.

$(5/2)x + 2/3 = -7/12$

$(5/2)x = -7/12 - 8/12 = -15/12 = -5/4$

$x = -5/4 \times 2/5 = -1/2$

The number is **−1/2**.

4 **A positive number is 5 times another number. If 21 is added to both the numbers, then one of the new numbers becomes twice the other new number. What are the numbers?**

✔ ANSWER

Let the numbers be x and 5x.

New numbers: x + 21 and 5x + 21. The bigger one is twice the smaller:

$5x + 21 = 2(x + 21)$

$5x + 21 = 2x + 42$

$3x = 21$, so $x = 7$

The numbers are **7 and 35**. (Check: 28 and 56, and $56 = 2 \times 28$ ✓)

5 **If you have ₹800 and you save ₹250 every month, find the amount you have after (i) 6 months (ii) 2 years. Express this as a linear pattern.**

✔ ANSWER

Amount after n months = 800 + 250n

(i) 800 + 250 × 6 = 800 + 1500 = **₹2300**

(ii) 2 years = 24 months: 800 + 250 × 24 = 800 + 6000 = **₹6800**

Linear pattern: 1050, 1300, 1550, … i.e. **800 + 250n** (increases by ₹250 every month).

***6** **The digits of a two-digit number differ by 3. If the digits are interchanged, and the resulting number is added to the original number, we get 143. Find both the numbers.**

✔ ANSWER

Let the tens digit be a and the units digit be b. Number = 10a + b, reversed number = 10b + a.

(10a + b) + (10b + a) = 143

11(a + b) = 143, so a + b = 13

Digits differ by 3: a − b = 3 (or b − a = 3)

Adding: 2a = 16 → a = 8, b = 5

The numbers are **85 and 58**. (Check: 85 + 58 = 143 ✓)

***7** **Draw the graph of the following equations, and identify their slopes and y-intercepts. Also, find the coordinates of the points where these lines cut the y-axis.**

(i) y = −3x + 4

(ii) 2y = 4x + 7

(iii) 5y = 6x − 10

(iv) 3y = 6x − 11

Are any of the lines parallel?

✔ ANSWER

Write each in the form y = ax + b.

(i) y = −3x + 4: slope **−3**, y-intercept **4**, cuts y-axis at **(0, 4)**. Points to plot: (0, 4), (1, 1), (2, −2).

(ii) y = 2x + 7/2: slope **2**, y-intercept **7/2**, cuts y-axis at **(0, 7/2)**. Points: (0, 3.5), (1, 5.5).

(iii) y = (6/5)x − 2: slope **6/5**, y-intercept **−2**, cuts y-axis at **(0, −2)**. Points: (0, −2), (5, 4).

(iv) y = 2x − 11/3: slope **2**, y-intercept **−11/3**, cuts y-axis at **(0, −11/3)**. Points: (0, −11/3), (2, 1/3).

Lines (ii) and (iv) have the same slope 2, so **(ii) and (iv) are parallel**.

***8** **y = 9/5 (x − 273) + 32, where x is in Kelvin and y in °F.**
(i) Find the temperature in Fahrenheit if it is 313 K.
(ii) If the temperature is 158 °F, find the temperature in Kelvin.

✔ ANSWER

(i) y = 9/5 × (313 − 273) + 32 = 9/5 × 40 + 32 = 72 + 32 = **104 °F**

(ii) 158 = 9/5 (x − 273) + 32

126 = 9/5 (x − 273)

x − 273 = 126 × 5/9 = 70

x = **343 K**

***9** **Work done = constant force × distance. Express this as a linear equation in w and d, and draw its graph taking the constant force as 3 units. What is the work done when the distance travelled is 2 units? Verify it by plotting it on the graph.**

✔ ANSWER

Equation: **w = 3d**

Points: (0, 0), (1, 3), (2, 6), (3, 9). Take d on the x-axis and w on the y-axis, plot and join: a straight line through the origin.

When d = 2: w = 3 × 2 = **6 units**.

On the graph, go up from d = 2 to the line and across to the w-axis: it reads 6 ✓

***10** **The graph of a linear polynomial p(x) passes through the points (1, 5) and (3, 11).**
(i) Find the polynomial p(x).
(ii) Find the coordinates where the graph of p(x) cuts the axes.
(iii) Draw the graph of p(x) and verify your answers.

✔ ANSWER

(i) Let p(x) = ax + b.

a + b = 5 and 3a + b = 11

Subtracting: 2a = 6 → a = 3, b = 2

p(x) = 3x + 2

(ii) y-axis (x = 0): p(0) = 2 → **(0, 2)**

x-axis (p(x) = 0): 3x + 2 = 0 → x = −2/3 → **(−2/3, 0)**

(iii) Plot (1, 5), (3, 11), (0, 2) and (−2/3, 0): all four lie on one straight line, which confirms the answers.

***11** **Let p(x) = ax + b and q(x) = cx + d be two linear polynomials such that:**

(i) p(0) = 5.

(ii) The polynomial p(x) − q(x) cuts the x-axis at (3, 0).

(iii) The sum p(x) + q(x) is equal to 6x + 4 for all real x.

Find the polynomials p(x) and q(x).

✔ ANSWER

From (i): p(0) = b = 5.

From (iii): (a + c)x + (b + d) = 6x + 4 → a + c = 6 and b + d = 4 → d = −1.

From (ii): p(x) − q(x) = (a − c)x + (b − d) = (a − c)x + 6, and it is 0 at x = 3:

3(a − c) + 6 = 0 → a − c = −2

a + c = 6 and a − c = −2 → a = 2, c = 4

p(x) = 2x + 5, q(x) = 4x − 1

Check: p − q = −2x + 6 = 0 at x = 3 ✓; p + q = 6x + 4 ✓

***12** **A growing pattern of hexagons made using matchsticks (Stage 1: 1 hexagon, Stage 2: 2, Stage 3: 3). Each new hexagon shares a side with the last hexagon.**

(i) Draw the next two stages. How many matchsticks will be required at these stages?

(ii) Complete the table: Stage number 1, 2, 3, 4, 5, ..., n; Number of matchsticks.

(iii) Find a rule for the number of matchsticks required for the nth stage.

(iv) How many matchsticks will be required for the 15th stage?

(v) Can 200 matchsticks form a stage in this pattern? Justify your answer.

✔ ANSWER

Stage 1 needs 6 sticks. Each new hexagon shares 1 side, so it needs only 5 more sticks.

(i) Stage 4: 4 hexagons in a chain → **21 matchsticks**; Stage 5: 5 hexagons → **26 matchsticks**.

(ii)

Stage number	1	2	3	4	5	...	n
Number of matchsticks	6	11	16	21	26	...	5n + 1

(iii) **Matchsticks at stage n = 5n + 1**

(iv) 5 × 15 + 1 = **76 matchsticks**

(v) 5n + 1 = 200 → 5n = 199 → n = 39.8, not a whole number. So **No**, 200 matchsticks cannot form a stage. (Stage 39 needs 196, stage 40 needs 201.)

***13** **Let p(x) = ax + b and q(x) = cx + d be two linear polynomials such that:**
(i) The graph of p(x) passes through the points (2, 3) and (6, 11).
(ii) The graph of q(x) passes through the point (4, −1).
(iii) The graph of q(x) is parallel to the graph of p(x).
Find p(x) and q(x). Also, find the coordinates of the points where these lines meet the x-axis.

✔ ANSWER

From (i): 2a + b = 3 and 6a + b = 11 → 4a = 8 → a = 2, b = 3 − 4 = −1

p(x) = 2x − 1

From (iii): parallel lines have the same slope, so c = 2.

From (ii): 2 × 4 + d = −1 → d = −9

q(x) = 2x − 9

x-axis: 2x − 1 = 0 → x = 1/2 → p meets it at **(1/2, 0)**

2x − 9 = 0 → x = 9/2 → q meets it at **(9/2, 0)**

***14** **What do all linear functions of the form f(x) = ax + a, a > 0, have in common?**

✔ ANSWER

f(x) = ax + a = a(x + 1)

- f(−1) = 0, so **all their graphs pass through the point (−1, 0)** on the x-axis.
- The slope and the y-intercept are equal (both = a).
- Since a > 0, all have a positive slope: all show **linear growth** (lines rise from left to right).

🎉 **Chapter 2 done! Revise the answers once more tomorrow.**